

SHARATH M

AI/ML Engineer | LLM & Generative AI Systems

+91-7353491465 | sharathparaiji@gmail.com | linkedin.com/in/sharath-m-62791a257 | github.com/sharathop

PROFESSIONAL SUMMARY

AI/ML Engineer with hands-on production experience across the ML and LLM lifecycle — data prep, feature engineering, model training/validation, and deployment. Shipped a phishing-URL detection model to AWS (46% fewer false negatives) and independently built a self-correcting RAG pipeline, a multi-agent research system, and an LLM evaluation framework. Skilled in Python, PyTorch, Scikit-learn, LangChain/LangGraph, and FastAPI, with a focus on reliable, evaluated AI systems.

EXPERIENCE

AI/ML Intern — Infosys Springboard (Remote)

Feb 2026 – Apr 2026

[Live Demo](#) | github.com/sharathop/phishing-detection-system

- Owned the full lifecycle of an AI/ML phishing-URL detection system in Python — data preparation through production deployment on AWS EC2 (Ubuntu) — as the sole engineer on the project.
- Engineered 56 lexical features (URL length, special characters, keyword patterns, suspicious TLDs, domain structure) from 11,430 balanced URLs using Pandas/NumPy, restricting inputs to URL structure to keep inference latency low and avoid live-content-fetch risk.
- Trained and benchmarked Logistic Regression, Decision Tree, Random Forest, and XGBoost in Scikit-learn; selected XGBoost and cut false negatives by 46% through threshold tuning and class-weighting, reaching 95.4% recall at 92.3% accuracy for a precision/recall trade-off suited to security use cases.
- Root-caused a silent train/inference feature mismatch discovered post-deployment — a drift-style failure invisible to offline validation — then patched and redeployed to restore prediction accuracy.
- Deployed the FastAPI service on AWS EC2 (Ubuntu) behind an Nginx reverse proxy as a systemd-managed service, connected to Neon PostgreSQL, and maintained releases via a Git-pull/systemctl-restart workflow.
- Built and shipped a Gemini-powered chatbot and analytics dashboard on top of the detection service, working directly with stakeholders to translate requirements into production features.

PROJECTS

Agentic Research & Report Generator — Multi-Agent Deep Research System

Python · LangGraph · Groq · Tavily · Streamlit

[Live Demo](#) | github.com/sharathop/agentic-research-report

- Architected a multi-agent research pipeline in LangGraph that decomposes broad research questions into 3–5 parallel sub-questions, fanning out independent agents to cut sequential query latency.
- Designed a self-critique and targeted-retry loop so each agent evaluates its own output and reissues a refined query on low-confidence answers, instead of surfacing incomplete research.
- Implemented isolated per-branch failure handling so one failed sub-agent doesn't block report generation, improving resilience over a naive sequential pipeline.
- Deployed the system on Streamlit Community Cloud with Tavily for live web retrieval and Groq for low-latency inference, exposing a working public demo end-to-end.

Self-Correcting RAG Pipeline — Closed-Loop Retrieval & Verification System

Python · LangChain · FAISS · BM25 · Cross-Encoder

[Live Demo](#) | github.com/sharathop/self-eval-rag

- Built a hybrid retrieval layer combining FAISS dense vector search with BM25 sparse retrieval, followed by cross-encoder reranking to improve top-k relevance over single-method retrieval.
- Designed a two-stage self-evaluation layer that scores factual and contextual grounding of generated answers, automatically triggering a retry when confidence falls below threshold.
- Added an LLM-judge override to the evaluation layer to catch cases where the automated grounding score would incorrectly reject a valid answer, reducing false rejections.
- Deployed the pipeline as a Streamlit app with a LangChain-orchestrated retrieval/generation flow for interactive end-to-end query testing.

LLM Evaluation & Hallucination Detection Framework — Model Evaluation & Explainability Tooling

Python · FastAPI · SQLite · Streamlit · HuggingFace

[Live Demo](#) | github.com/sharathop/llm-eval-framework

- Built a multi-metric evaluation engine scoring LLM outputs on NLI-based factual consistency, BERTScore, cosine relevance, and fluency, plus RAG-specific context relevance and recall.
- Designed the framework to be model-agnostic, so outputs from any LLM can be scored against the same metric suite for apples-to-apples comparison.
- Built a FastAPI backend with SQLite persistence to store evaluation runs, paired with a Streamlit dashboard for exploring results and drilling into individual metric scores.
- Deployed the tool on HuggingFace Spaces as a standalone evaluation utility usable outside the original development environment.

TECHNICAL SKILLS

Programming: Python · SQL · Java

Machine Learning: Scikit-learn · PyTorch · XGBoost · Transformers · CNNs · Pandas · NumPy · EDA · Feature Engineering · Model Training & Validation · Model Monitoring

LLM & Generative AI: LangChain · LangGraph · Multi-Agent Orchestration · RAG Pipelines · FAISS · BM25 · Cross-Encoder Reranking · HuggingFace · Prompt Engineering · LLM Evaluation & Hallucination Detection

Backend: FastAPI · REST APIs · Streamlit

Cloud & Infra: AWS EC2 · Nginx · systemd · Ubuntu Linux · HuggingFace Spaces · Streamlit Community Cloud · Vercel · Render

Databases: PostgreSQL (Neon) · SQLite

DevOps & Tools: Git · GitHub · Jupyter · VS Code

EDUCATION

Master of Computer Applications (MCA) — Maharaja Institute of Technology, Thandavapura, Mysore, KA | 2024–2026 · CGPA: 8.9/10

Bachelor of Computer Applications (BCA) — JSS College of Arts, Commerce and Science, Mysore, KA | 2021–2024 · CGPA: 8.02/10

ADDITIONAL

Languages: English (Professional) · Kannada (Native) · Hindi (Conversational) **Interests:** Open-Source Software · Cricket · eSports